

AERIS Expert Review Packet

Anomaly

Source / metric	tceq / so2
Observed / expected	3.1 / -0.484828
Time	2026-07-19 16:00 UTC
Location	29.6240, -95.4743
Severity	severe
Detectors	zscore, stl, isolation_forest

Stored evidence summary

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	humidity	percent	9 / 840	37.3 to 100; mean 74.9209	70.35 at 2026-07-19 15:55Z; dt 5 min
asos	temperature	degC	9 / 840	22 to 37; mean 29.4469	30 at 2026-07-19 15:55Z; dt 5 min
asos	wind_direction	degree	9 / 1120	0 to 320; mean 160.071	200 at 2026-07-19 15:55Z; dt 5 min
asos	wind_speed	m/s	9 / 1165	0 to 8.74555; mean 2.4283	2.05778 at 2026-07-19 15:55Z; dt 5 min
noaa_gfs	gh_500	m	11 / 132	5918.94 to 5946.98; mean 5930.45	5926.69 at 2026-07-19 18:00Z; dt 120 min
noaa_gfs	pbl_height	m	11 / 132	64.8946 to 1860.78; mean 655.241	1356.63 at 2026-07-19 18:00Z; dt 120 min
noaa_gfs	precipitable_water	mm	11 / 132	29.224 to 47.063; mean 36.6002	36.0176 at 2026-07-19 18:00Z; dt 120 min
noaa_gfs	surface_pressure	hPa	11 / 132	1004.97 to 1018.53; mean 1012.58	1013.14 at 2026-07-19 18:00Z; dt 120 min
noaa_gfs	t_850	degC	11 / 132	17.9655 to 22.9209; mean 21.4415	22.0615 at 2026-07-19 18:00Z; dt 120 min
noaa_gfs	u_10m	m/s	11 / 132	-2.63161 to 2.90839; mean 0.7456	1.13125 at 2026-07-19 18:00Z; dt 120 min
noaa_gfs	v_10m	m/s	11 / 132	-1.04667 to 4.7729; mean 2.1815	1.9878 at 2026-07-19 18:00Z; dt 120 min
openaq	ozone	ppm	16 / 1053	-0.004 to 0.08; mean 0.0182	0.018 at 2026-07-19 16:00Z; dt 0 min
openaq	pm10	ug/m3	2 / 143	29 to 66; mean 41.7413	32 at 2026-07-19 16:00Z; dt 0 min
openaq	pm25	ug/m3	76 / 3227	0.3 to 45.6; mean 7.7314	4.7 at 2026-07-19 16:00Z; dt 0 min
openweather	cloud_cover	percent	5 / 357	0 to 100; mean 57.8739	92 at 2026-07-19 16:01Z; dt 1.9 min
openweather	humidity	percent	5 / 357	44 to 96; mean 76.1793	69 at 2026-07-19 16:01Z; dt 1.9 min
openweather	precipitation	mm/h	5 / 357	0 to 0; mean 0	0 at 2026-07-19 16:01Z; dt 1.9 min
openweather	pressure	hPa	5 / 357	1011 to 1020; mean 1015.38	1017 at 2026-07-19 16:01Z; dt 1.9 min
openweather	temperature	degC	5 / 357	24.26 to 37.22; mean 30.2348	31.89 at 2026-07-19 16:01Z; dt 1.9 min
openweather	wind_direction	degree	5 / 357	0 to 336; mean 195.541	186 at 2026-07-19 16:01Z; dt 1.9 min
openweather	wind_speed	m/s	5 / 357	0.45 to 5.9; mean 2.0308	0.45 at 2026-07-19 16:01Z; dt 1.9 min
purpleair	pm25	ug/m3	72 / 5081	0 to 83.5; mean 9.2691	8.3 at 2026-07-19 16:00Z; dt 0 min
sentinel5p	s5p_aer_ai_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-07-19 19:42Z; dt 222.9 min

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
sentinel5p	s5p_ch4_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-07-19 19:42Z; dt 222.9 min
sentinel5p	s5p_co_column	mol/m^2	4 / 4	0.0254212 to 0.0262849; mean 0.0259	0.0255937 at 2026-07-19 19:42Z; dt 222.9 min
sentinel5p	s5p_co_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-07-19 19:42Z; dt 222.9 min
sentinel5p	s5p_hcho_column	mol/m^2	7 / 7	0.000157838 to 0.000213688; mean 0.0002	0.000176599 at 2026-07-19 20:09Z; dt 249.6 min
sentinel5p	s5p_hcho_granule_available	count	7 / 7	1 to 1; mean 1	1 at 2026-07-19 20:09Z; dt 249.6 min
sentinel5p	s5p_no2_column	mol/m^2	4 / 4	1.64557e-05 to 5.83485e-05; mean 0	3.22877e-05 at 2026-07-19 20:09Z; dt 249.6 min
sentinel5p	s5p_no2_granule_available	count	4 / 4	1 to 1; mean 1	1 at 2026-07-19 20:09Z; dt 249.6 min
sentinel5p	s5p_o3_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-07-19 19:42Z; dt 222.9 min
sentinel5p	s5p_so2_column	mol/m^2	7 / 7	-3.60822e-05 to 0.000116366; mean 0	2.99088e-06 at 2026-07-19 20:09Z; dt 249.6 min
sentinel5p	s5p_so2_granule_available	count	7 / 7	1 to 1; mean 1	1 at 2026-07-19 20:09Z; dt 249.6 min
tceq	co	ppm	3 / 182	0.1 to 0.8; mean 0.2187	0.3 at 2026-07-19 16:00Z; dt 0 min
tceq	no2	ppb	12 / 680	-2.5 to 20.7; mean 3.8793	0.8 at 2026-07-19 16:00Z; dt 0 min
tceq	so2	ppb	3 / 176	-0.6 to 6.9; mean 0.0915	3.1 at 2026-07-19 16:00Z; dt 0 min

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

Source	Metric	Values
asos	humidity	07-18 00Z 86.82, 06Z 90.86, 12Z 73.25, 18Z 58.09, 07-19 00Z 80.11, 06Z 93.72, 12Z 75.18, 18Z 54.34, 07-20 00Z 79.67, 06Z 94.44, 12Z 77.08, 18Z 48.96, 07-21 00Z 72.44
asos	temperature	07-18 00Z 27.19, 06Z 26.48, 12Z 30.36, 18Z 33.1, 07-19 00Z 28.26, 06Z 25.26, 12Z 29.45, 18Z 33.97, 07-20 00Z 28.69, 06Z 24.95, 12Z 28.8, 18Z 34.38, 07-21 00Z 30.04
asos	wind_direction	07-18 00Z 143.8, 06Z 120, 12Z 176, 18Z 182.5, 07-19 00Z 166.2, 06Z 81.77, 12Z 186.7, 18Z 139.9, 07-20 00Z 167.3, 06Z 133.3, 12Z 225.3, 18Z 176.1, 07-21 00Z 169.5
asos	wind_speed	07-18 00Z 2.466, 06Z 1.905, 12Z 2.761, 18Z 3.809, 07-19 00Z 3.061, 06Z 0.926, 12Z 2.361, 18Z 2.239, 07-20 00Z 2.895, 06Z 1.554, 12Z 2.562, 18Z 2.234, 07-21 00Z 2.907
noaa_gfs	gh_500	07-18 06Z 5946, 12Z 5938, 18Z 5942, 07-19 00Z 5929, 06Z 5935, 12Z 5923, 18Z 5927, 07-20 00Z 5922, 06Z 5931, 12Z 5920, 18Z 5929, 07-21 00Z 5924
noaa_gfs	pbl_height	07-18 06Z 392.8, 12Z 268.8, 18Z 1591, 07-19 00Z 757.2, 06Z 284.4, 12Z 164.6, 18Z 1354, 07-20 00Z 691.3, 06Z 222.8, 12Z 142, 18Z 1254, 07-21 00Z 739.6
noaa_gfs	precipitable_water	07-18 06Z 39.17, 12Z 40.6, 18Z 43.53, 07-19 00Z 37.63, 06Z 36.82, 12Z 34.83, 18Z 36.18, 07-20 00Z 38.41, 06Z 33.19, 12Z 29.95, 18Z 32.96, 07-21 00Z 35.91
noaa_gfs	surface_pressure	07-18 06Z 1016, 12Z 1016, 18Z 1015, 07-19 00Z 1012, 06Z 1013, 12Z 1013, 18Z 1013, 07-20 00Z 1010, 06Z 1011, 12Z 1011, 18Z 1011, 07-21 00Z 1008
noaa_gfs	t_850	07-18 06Z 21.06, 12Z 20.78, 18Z 18.57, 07-19 00Z 20.78, 06Z 22.45, 12Z 21.95, 18Z 21.53, 07-20 00Z 20.98, 06Z 22.73, 12Z 22.67, 18Z 22.38, 07-21 00Z 21.42
noaa_gfs	u_10m	07-18 06Z -0.6589, 12Z 0.49, 18Z 0.4517, 07-19 00Z 0.153, 06Z 1.176, 12Z 0.6183, 18Z 1.386, 07-20 00Z 0.05235, 06Z 1.324, 12Z 1.67, 18Z 2.052, 07-21 00Z 0.2338
noaa_gfs	v_10m	07-18 06Z 3.108, 12Z 2.039, 18Z 3.212, 07-19 00Z 3.971, 06Z 2.266, 12Z 1.697, 18Z 1.66, 07-20 00Z 2.493, 06Z 2.464, 12Z 0.6489, 18Z 0.8407, 07-21 00Z 1.78

Source	Metric	Values
openaq	ozone	07-18 00Z 0.01384, 06Z 0.01101, 12Z 0.01498, 18Z 0.02526, 07-19 00Z 0.01543, 06Z 0.008361, 12Z 0.01367, 18Z 0.03243, 07-20 00Z 0.01721, 06Z 0.006975, 12Z 0.01064, 18Z 0.04057, 07-21 00Z 0.02118
openaq	pm10	07-18 00Z 35, 06Z 34.36, 12Z 39.5, 18Z 37.18, 07-19 00Z 39.33, 06Z 35.58, 12Z 35.75, 18Z 46.83, 07-20 00Z 45.25, 06Z 44.75, 12Z 49.67, 18Z 50.75, 07-21 00Z 43.67
openaq	pm25	07-18 00Z 8.073, 06Z 8.918, 12Z 8.387, 18Z 5.677, 07-19 00Z 7.423, 06Z 8.501, 12Z 6.148, 18Z 8.112, 07-20 00Z 7.621, 06Z 8.657, 12Z 7.838, 18Z 6.568, 07-21 00Z 6.504
openweather	cloud_cover	07-18 00Z 85.5, 06Z 96.72, 12Z 84.16, 18Z 94.1, 07-19 00Z 94.31, 06Z 93.58, 12Z 94.52, 18Z 84.9, 07-20 00Z 12.83, 06Z 5.645, 12Z 3.833, 18Z 2.967, 07-21 00Z 4.15
openweather	humidity	07-18 00Z 84.1, 06Z 89.38, 12Z 79.68, 18Z 63.13, 07-19 00Z 77.14, 06Z 90.48, 12Z 81.63, 18Z 60.1, 07-20 00Z 74.79, 06Z 90.77, 12Z 80.67, 18Z 53.33, 07-21 00Z 67.35
openweather	precipitation	07-18 00Z 0, 06Z 0, 12Z 0, 18Z 0, 07-19 00Z 0, 06Z 0, 12Z 0, 18Z 0, 07-20 00Z 0, 06Z 0, 12Z 0, 18Z 0, 07-21 00Z 0
openweather	pressure	07-18 00Z 1019, 06Z 1018, 12Z 1019, 18Z 1016, 07-19 00Z 1016, 06Z 1016, 12Z 1017, 18Z 1014, 07-20 00Z 1013, 06Z 1014, 12Z 1015, 18Z 1012, 07-21 00Z 1012
openweather	temperature	07-18 00Z 27.97, 06Z 26.98, 12Z 29.79, 18Z 34.26, 07-19 00Z 29.62, 06Z 26.2, 12Z 28.9, 18Z 35.34, 07-20 00Z 30.62, 06Z 25.88, 12Z 28.74, 18Z 35.79, 07-21 00Z 32.12
openweather	wind_direction	07-18 00Z 182.3, 06Z 180.2, 12Z 194.5, 18Z 174.5, 07-19 00Z 183.4, 06Z 192.7, 12Z 213.6, 18Z 184.9, 07-20 00Z 186.7, 06Z 194.4, 12Z 239.7, 18Z 215.6, 07-21 00Z 189.6
openweather	wind_speed	07-18 00Z 1.868, 06Z 1.972, 12Z 2.012, 18Z 1.928, 07-19 00Z 2.57, 06Z 1.94, 12Z 1.676, 18Z 1.968, 07-20 00Z 2.389, 06Z 1.895, 12Z 1.845, 18Z 1.766, 07-21 00Z 2.68
purpleair	pm25	07-18 00Z 9.86, 06Z 10.92, 12Z 9.609, 18Z 7.344, 07-19 00Z 8.974, 06Z 10.5, 12Z 8.618, 18Z 8.636, 07-20 00Z 9.265, 06Z 10.82, 12Z 9.817, 18Z 7.625, 07-21 00Z 8.797
sentinel5p	s5p_aer_ai_granule_available	07-18 18Z 1, 07-19 18Z 1, 07-20 18Z 1
sentinel5p	s5p_ch4_granule_available	07-18 18Z 1, 07-19 18Z 1, 07-20 18Z 1
sentinel5p	s5p_co_column	07-19 18Z 0.02551, 07-20 18Z 0.02628
sentinel5p	s5p_co_granule_available	07-18 18Z 1, 07-19 18Z 1, 07-20 18Z 1
sentinel5p	s5p_hcho_column	07-18 18Z 0.0001745, 07-19 18Z 0.0001766, 07-20 18Z 0.000213
sentinel5p	s5p_hcho_granule_available	07-18 18Z 1, 07-19 18Z 1, 07-20 18Z 1
sentinel5p	s5p_no2_column	07-18 18Z 1.872e-05, 07-19 18Z 3.229e-05, 07-20 18Z 5.835e-05
sentinel5p	s5p_no2_granule_available	07-18 18Z 1, 07-19 18Z 1, 07-20 18Z 1
sentinel5p	s5p_o3_granule_available	07-18 18Z 1, 07-19 18Z 1, 07-20 18Z 1
sentinel5p	s5p_so2_column	07-18 18Z 3.278e-05, 07-19 18Z 2.991e-06, 07-20 18Z 1.73e-05
sentinel5p	s5p_so2_granule_available	07-18 18Z 1, 07-19 18Z 1, 07-20 18Z 1
tceq	co	07-18 06Z 0.2111, 12Z 0.2556, 18Z 0.2556, 07-19 00Z 0.25, 06Z 0.1944, 12Z 0.2111, 18Z 0.2, 07-20 00Z 0.2667, 06Z 0.2056, 12Z 0.2125, 18Z 0.1722, 07-21 00Z 0.2222
tceq	no2	07-18 06Z 3.628, 12Z 3.301, 18Z 2.128, 07-19 00Z 1.933, 06Z 3.03, 12Z 2.541, 18Z 3.297, 07-20 00Z 3.545, 06Z 4.577, 12Z 6.21, 18Z 6.136, 07-21 00Z 6.723
tceq	so2	07-18 06Z -0.3611, 12Z -0.3222, 18Z -0.2889, 07-19 00Z -0.1, 06Z -0.4533, 12Z 0.1056, 18Z 0.4389, 07-20 00Z -0.2875, 06Z -0.3471, 12Z -0.175, 18Z 1.906, 07-21 00Z 1.213

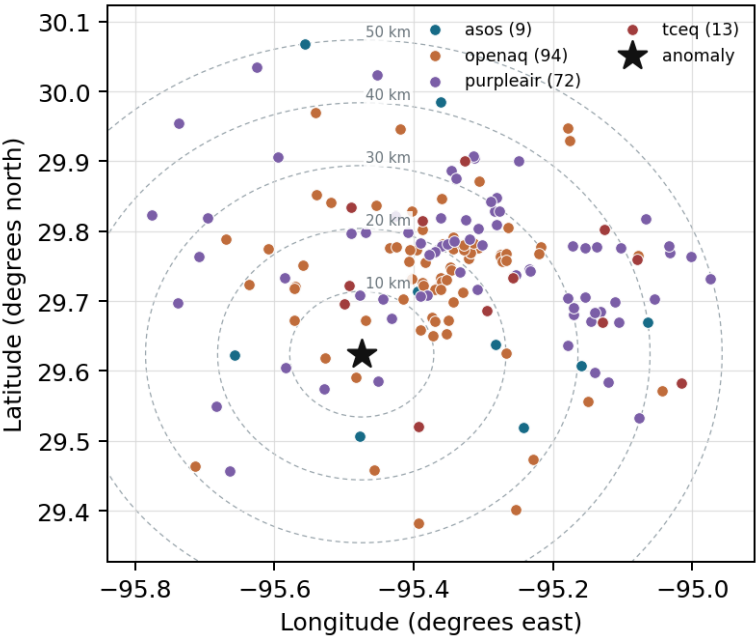
Per-site averages

Each metric averaged at each site, with that site's distance from the event in brackets.

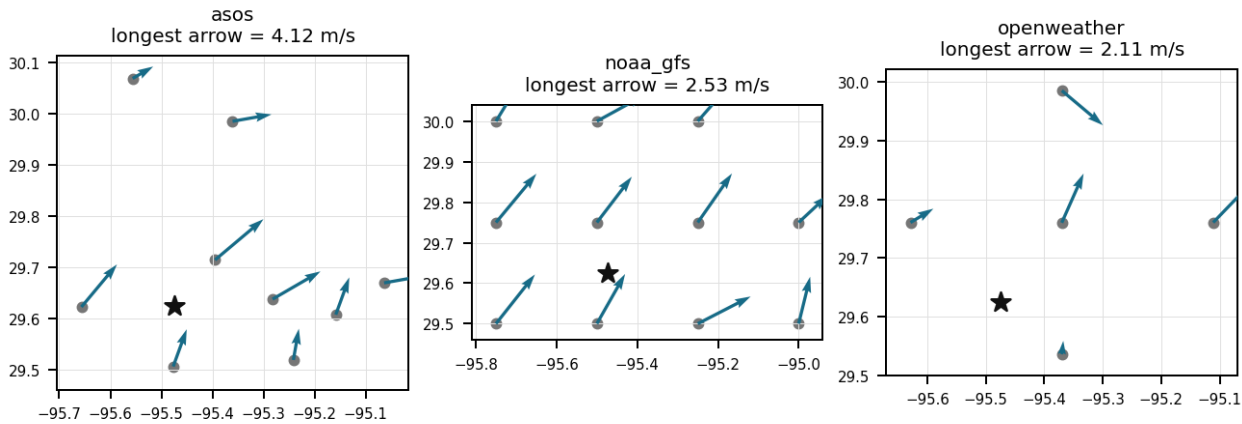
Source	Metric	Values
asos	humidity	MCJ (12.608 km) 72.43, AXH (13.11 km) 72.48, SGR (17.608 km) 79.95, HOU (18.613 km) 74.61, LVJ (25.353 km) 75.27, EFD (30.569 km) 74.76, T41 (39.963 km) 74.99, IAH (41.55 km) 73.28, +1 more
asos	temperature	MCJ (12.608 km) 29.83, AXH (13.11 km) 29.39, SGR (17.608 km) 28.71, HOU (18.613 km) 29.7, LVJ (25.353 km) 29.32, EFD (30.569 km) 29.86, T41 (39.963 km) 29.84, IAH (41.55 km) 29.99, +1 more
asos	wind_direction	MCJ (12.608 km) 170.6, AXH (13.11 km) 103.9, SGR (17.608 km) 173.3, HOU (18.613 km) 192.6, LVJ (25.353 km) 134.3, EFD (30.569 km) 169, T41 (39.963 km) 185, IAH (41.55 km) 167.9, +1 more
asos	wind_speed	MCJ (12.608 km) 3.252, AXH (13.11 km) 0.9825, SGR (17.608 km) 2.726, HOU (18.613 km) 3.155, LVJ (25.353 km) 1.786, EFD (30.569 km) 3.06, T41 (39.963 km) 3.31, IAH (41.55 km) 2.244, +1 more
noaa_gfs	gh_500	gfs:29.50,-95.50 (14.007 km) 5931, gfs:29.75,-95.50 (14.23 km) 5930, gfs:29.50,-95.25 (25.708 km) 5931, gfs:29.75,-95.25 (25.808 km) 5931, gfs:29.50,-95.75 (30.014 km) 5931, gfs:29.75,-95.75 (30.09 km) 5930, gfs:30.00,-95.50 (41.885 km) 5930, gfs:30.00,-95.25 (47.082 km) 5930, +3 more
noaa_gfs	pbl_height	gfs:29.50,-95.50 (14.007 km) 499.5, gfs:29.75,-95.50 (14.23 km) 903, gfs:29.50,-95.25 (25.708 km) 480.1, gfs:29.75,-95.25 (25.808 km) 789.9, gfs:29.50,-95.75 (30.014 km) 465.6, gfs:29.75,-95.75 (30.09 km) 581.5, gfs:30.00,-95.50 (41.885 km) 902.2, gfs:30.00,-95.25 (47.082 km) 864.3, +3 more
noaa_gfs	precipitable_water	gfs:29.50,-95.50 (14.007 km) 36.77, gfs:29.75,-95.50 (14.23 km) 36.37, gfs:29.50,-95.25 (25.708 km) 37.29, gfs:29.75,-95.25 (25.808 km) 37.19, gfs:29.50,-95.75 (30.014 km) 35.88, gfs:29.75,-95.75 (30.09 km) 35.44, gfs:30.00,-95.50 (41.885 km) 36, gfs:30.00,-95.25 (47.082 km) 36.89, +3 more
noaa_gfs	surface_pressure	gfs:29.50,-95.50 (14.007 km) 1013, gfs:29.75,-95.50 (14.23 km) 1012, gfs:29.50,-95.25 (25.708 km) 1014, gfs:29.75,-95.25 (25.808 km) 1014, gfs:29.50,-95.75 (30.014 km) 1012, gfs:29.75,-95.75 (30.09 km) 1011, gfs:30.00,-95.50 (41.885 km) 1011, gfs:30.00,-95.25 (47.082 km) 1012, +3 more
noaa_gfs	t_850	gfs:29.50,-95.50 (14.007 km) 21.43, gfs:29.75,-95.50 (14.23 km) 21.2, gfs:29.50,-95.25 (25.708 km) 21.56, gfs:29.75,-95.25 (25.808 km) 21.41, gfs:29.50,-95.75 (30.014 km) 21.53, gfs:29.75,-95.75 (30.09 km) 21.51, gfs:30.00,-95.50 (41.885 km) 21.23, gfs:30.00,-95.25 (47.082 km) 21.12, +3 more
noaa_gfs	u_10m	gfs:29.50,-95.50 (14.007 km) 0.4518, gfs:29.75,-95.50 (14.23 km) 0.9068, gfs:29.50,-95.25 (25.708 km) 0.491, gfs:29.75,-95.25 (25.808 km) 0.6793, gfs:29.50,-95.75 (30.014 km) 0.9568, gfs:29.75,-95.75 (30.09 km) 1.023, gfs:30.00,-95.50 (41.885 km) 1.325, gfs:30.00,-95.25 (47.082 km) 0.4668, +3 more
noaa_gfs	v_10m	gfs:29.50,-95.50 (14.007 km) 2.421, gfs:29.75,-95.50 (14.23 km) 2.099, gfs:29.50,-95.25 (25.708 km) 2.409, gfs:29.75,-95.25 (25.808 km) 2.203, gfs:29.50,-95.75 (30.014 km) 2.107, gfs:29.75,-95.75 (30.09 km) 2.129, gfs:30.00,-95.50 (41.885 km) 1.678, gfs:30.00,-95.25 (47.082 km) 1.708, +3 more
openaq	ozone	312 (0.017 km) 0.01762, 311 (8.339 km) 0.01837, 3055 (13.984 km) 0.01435, 1319333 (18.217 km) 0.02245, 325 (18.693 km) 0.004938, 287 (19.112 km) 0.01876, 320 (20.024 km) 0.01554, 319 (23.419 km) 0.01796, +8 more
openaq	pm10	15852419 (22.104 km) 36.51, 271 (33.811 km) 47.2
openaq	pm25	13955814 (3.744 km) 4.703, 15399615 (5.167 km) 4.866, 8967119 (5.456 km) 11.14, 14140752 (8.986 km) 8.615, 8967021 (10.262 km) 9.59, 8967001 (10.484 km) 6.46, 13955816 (10.805 km) 9.451, 15539683 (11.385 km) 8.711, +68 more
openweather	cloud_cover	grid:south (14.084 km) 57.74, grid:center (18.176 km) 56.39, grid:west (21.256 km) 56.19, grid:east (38.226 km) 58.76, grid:north (41.392 km) 60.31
openweather	humidity	grid:south (14.084 km) 75.81, grid:center (18.176 km) 74.63, grid:west (21.256 km) 75.35, grid:east (38.226 km) 80.32, grid:north (41.392 km) 74.8
openweather	precipitation	grid:south (14.084 km) 0, grid:center (18.176 km) 0, grid:west (21.256 km) 0, grid:east (38.226 km) 0, grid:north (41.392 km) 0
openweather	pressure	grid:south (14.084 km) 1015, grid:center (18.176 km) 1015, grid:west (21.256 km) 1015, grid:east (38.226 km) 1015, grid:north (41.392 km) 1015
openweather	temperature	grid:south (14.084 km) 30, grid:center (18.176 km) 30.74, grid:west (21.256 km) 30.22, grid:east (38.226 km) 29.9, grid:north (41.392 km) 30.32
openweather	wind_direction	grid:south (14.084 km) 197.8, grid:center (18.176 km) 211.6, grid:west (21.256 km) 190.1, grid:east (38.226 km) 198.7, grid:north (41.392 km) 179.6
openweather	wind_speed	grid:south (14.084 km) 2.328, grid:center (18.176 km) 1.783, grid:west (21.256 km) 1.623, grid:east (38.226 km) 2.636, grid:north (41.392 km) 1.785

Source	Metric	Values
purpleair	pm25	161003 (4.856 km) 5.355, 166451 (7.01 km) 10.41, 161015 (7.666 km) 10.68, 46237 (9.249 km) 0.4781, 186315 (9.424 km) 10.18, 276512 (10.819 km) 10.9, 293735 (12.303 km) 11.23, 293725 (13.024 km) 10.66, +64 more
tceq	co	482011052 (22.762 km) 0.3651, 482011035 (24.236 km) 0.1322, 482011039 (33.805 km) 0.15
tceq	no2	482010055 (8.334 km) 2.154, 482011066 (11.0 km) 4.546, 480391004 (13.97 km) 1.677, 482010416 (18.688 km) 4.636, 482011052 (22.762 km) 7.646, 482010047 (23.42 km) 3.771, 482011035 (24.236 km) 5.603, 482011039 (33.805 km) 2.633, +4 more
tceq	so2	482010051 (0.0 km) 0.2767, 482011035 (24.236 km) -0.2534, 482011039 (33.805 km) 0.2448

Ground observation locations in the stored 72-hour context
dashed rings are great-circle distance from the anomaly



Event-nearest wind vectors from stored values (arrow points downwind)



Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). The labeling guide that comes with this packet has the definitions and marking rules. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Claim 1 of 35

The air quality anomaly is related to PM2.5 levels.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 2 of 35

Weak southerly to south-southwesterly low-level flow and only moderate surface wind speeds likely increased the probability that a nearby emission plume intersected the monitor without rapid dilution at 2026-07-19 16:00 UTC.

Calm-wind flag: wind direction unstable under calm conditions. openweather: event wind 0.45 m/s, below its cutoff of 1.5 m/s (mean - 2*SD gave 0.03549610439032991 m/s; n=357 wind readings). Cutoff floor: confirmed 2026-07-24.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 3 of 35

The PM2.5 levels are elevated at 8.3 ug/m3, which is higher than the mean of 9.2691 ug/m3.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 4 of 35

The air quality anomaly in Houston, Texas is caused by high levels of PM2.5.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 5 of 35

A ground-level SO2 spike reaching 3.1 ppb was recorded at TCEQ monitor 482010051 in Houston at 16:00 UTC on July 19, 2026.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 6 of 35

The anomalous sulfur dioxide measurement occurred at TCEQ monitor 482010051 in the Houston area, where the monitor mean over the context window was 0.2767 ppb.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 7 of 35

A short-duration plume from nearby industrial sulfur-emitting sources such as refinery, petrochemical, sulfur recovery, storage, or ship-channel operations is the leading physical cause of the 3.1 ppb SO₂ spike because the anomaly was highly localized at one TCEQ monitor while concurrent NO₂, CO, PM_{2.5}, PM₁₀, and ozone near the site were not comparably elevated.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 8 of 35

Industrial point-source emissions or process upset events at upwind petrochemical facilities near Houston likely caused the elevated SO₂ concentration detected by the ground station.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 9 of 35

NOAA GFS model output showed planetary boundary layer height expanding to over 1350 meters by 18:00 UTC on July 19, 2026, facilitating vertical dilution following early morning accumulation.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 10 of 35

A nearby industrial facility or power plant is releasing excessive amounts of particulate matter and volatile organic compounds into the atmosphere.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 11 of 35

The air quality anomaly in Houston, Texas is not caused by changes in temperature.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 12 of 35

The anomaly occurred on July 19th at approximately 16:01:53 UTC.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 13 of 35

The weather conditions on July 19th, including high pressure and light winds, contributed to the air quality anomaly in Houston, Texas.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 14 of 35

A severe sulfur dioxide (SO₂) anomaly occurred on July 19, 2026, at 16:00 UTC at coordinates (29.624, -95.474), reaching an observed concentration of 3.1 ppb compared to an expected value of -0.485 ppb, resulting in a z-score of 30.03.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 15 of 35

Surface wind speeds near the Houston anomaly location averaged approximately 2 m/s from the south to south-southwest during the spike event on July 19, 2026.

Calm-wind flag: wind direction unstable under calm conditions. openweather: event wind 0.45 m/s, below its cutoff of 1.5 m/s (mean - 2*SD gave 0.03549610439032991 m/s; n=357 wind readings). Cutoff floor: confirmed 2026-07-24.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 16 of 35

ASOS weather stations and OpenWeather reported light surface winds around 2 m/s from the south-southwest near the time of the SO₂ anomaly, indicating local plume dispersion from upwind industrial facilities.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 17 of 35

A monitor-specific measurement artifact or ultra-local nonregional SO₂ source remains a plausible alternative explanation because Sentinel-5P did not show a matching broad SO₂ column enhancement over the area later that afternoon.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 18 of 35

The temperature inversion over the Houston area is causing a buildup of pollutants near the surface.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 19 of 35

A monitor-specific artifact remains a viable alternative because the provided SO₂ network includes only three TCEQ monitors within 50 km and the two other SO₂ monitors have different site means and no explicit simultaneous co-spike reported at 2026-07-19 16:00 UTC, so independent spatial corroboration for the exact event is limited in the current dataset.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 20 of 35

The air quality anomaly in Houston, Texas is not caused by wind direction or speed.

Mark exactly one:

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Note:

Claim 21 of 35

The TCEQ monitor at 29.624, -95.474 measured 3.1 ppb SO₂ at 2026-07-19 16:00 UTC, which was far above the expected value of about -0.485 ppb and had a severe z-score of about 30, indicating an extreme local SO₂ excursion at that site.

Mark exactly one:

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Note:

Claim 22 of 35

Low surface wind speeds and southerly wind directions facilitated the transport of SO₂ plumes from upwind industrial sources toward the TCEQ monitoring location.

Mark exactly one:

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Note:

Claim 23 of 35

Regional supporting data did not show a broad sulfur pollution episode, because Sentinel-5P SO₂ near the area later that day was low, nearby NO₂ and CO were not elevated at the anomaly time, and nearby PM_{2.5} and ozone remained modest, so the most likely cause is a localized industrial sulfur release rather than region-wide accumulation or wildfire smoke.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 24 of 35

The pollutant showing the anomaly is sulfur dioxide measured by a TCEQ monitor.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 25 of 35

The air quality anomaly in Houston, Texas on July 19th was characterized by elevated levels of PM_{2.5}.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 26 of 35

The SO₂ concentration spike was localized to TCEQ monitor 482010051 at the anomaly location, whereas monitors 24.2 km and 33.8 km away maintained period averages of -0.2534 ppb and 0.2448 ppb, respectively.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 27 of 35

Sentinel-5P measured an SO₂ column density of 2.99×10^{-6} mol/m² near the anomaly site approximately four hours after the surface spike at 20:09:34 UTC.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 28 of 35

NO₂ levels were also elevated during the air quality anomaly in Houston, Texas on July 19th.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 29 of 35

Surface meteorology near the anomaly time showed weak southerly to south-southwesterly flow, with nearby observations around 0.45 to 2.06 m/s and wind directions near 186 to 200 degrees, which would favor transport of a nearby sulfur plume to the monitor with only modest dispersion.

Calm-wind flag: wind direction unstable under calm conditions. openweather: event wind 0.45 m/s, below its cutoff of 1.5 m/s (mean - 2*SD gave 0.03549610439032991 m/s; n=357 wind readings). Cutoff floor: confirmed 2026-07-24.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 30 of 35

A local industrial sulfur plume is supported because the TCEQ SO₂ monitor at the anomaly site measured 3.1 ppb at 2026-07-19 16:00 UTC against an expected value near -0.48 ppb and a monitor mean of 0.28 ppb, while nearby surface winds were weak and southerly to south-southwesterly at about 0.45 to 2.06 m/s, conditions consistent with limited dilution and transport from the Houston Ship Channel or other industrial sources to the north-northeast of air parcels arriving from the south.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 31 of 35

The wind direction and speed are not effectively dispersing pollutants, leading to a concentration of air pollutants in the Houston area.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 32 of 35

Boundary layer dynamics and weak atmospheric ventilation limited the dispersion of emitted sulfur dioxide, leading to localized ground-level accumulation.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 33 of 35

Ground monitoring recorded a sharp SO₂ concentration spike of 3.1 ppb at station 482010051 at 16:00 UTC on July 19, 2026, while Sentinel-5P column observations showed baseline atmospheric SO₂ levels.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 34 of 35

The TCEQ sulfur dioxide concentration at 29.624, -95.474 was 3.1 ppb at 2026-07-19T16:00:00+00:00, compared with an expected value of -0.48482758620689664 ppb and a z-score of 30.02768750590938, which is labeled severe.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 35 of 35

A broad regional sulfur event is contradicted because Sentinel-5P showed a low SO₂ column near the area later on 2026-07-19 at about 2.99e-06 mol/m², while satellite CO and HCHO were not elevated and surface PM_{2.5} near the anomaly time remained modest at about 4.7 to 8.3 ug/m³, which is not the signature of a large area-wide combustion or volcanic sulfur episode.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.
